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202. Supplementary Remarks on the Porism of the In-and-Circumscribed Triangle

[From the *Philosophical Magazine*, vol. XIII. (1857), pp. 19–30.]

In my former papers (see *Phil. Mag.* August and November, 1853, [115, 116]) I established (as part of a more general one) the following theorem, viz. the condition that there may be inscribed in the conic $U = 0$, an infinity of triangles circumscribed about the conic $V = 0$, is, that if we develop in ascending powers of k the square root of the discriminant of $kU + V$, the coefficient of k^2 in this development must vanish. Thus writing

$$\text{disct.} .(kV + U) = (K, \Theta, \Theta', K^{\frac{1}{2}}k, 1)^3,$$

the condition in question is found to be

$$3\Theta^2 - 4K\Theta' = 0.$$

The following investigations, although relating only to particular cases of the theorem, are, I think, not without interest.

If the equation of the conic containing the angles is

$$U = 2ayz + 2bzx + 2cxy = 0,$$

and the equation of the conic touched by the sides is

$$V = x^2 + y^2 + z^2 - 2yz - 2zx - 2xy = 0,$$

we have

$$\text{disct.} .(k, k, k, a - k, b - k, c - k \mid x, y, z)^2 = (K, \Theta, \Theta', K^{\frac{1}{2}}k, 1)^3,$$

that is,

$$\begin{aligned} K &= -4, \\ \Theta &= \frac{4}{3}(a + b + c), \\ \Theta' &= -\frac{4}{3}(a + b + c)^2, \\ K' &= 2abc. \end{aligned}$$

and the equation $3\Theta^2 - 4K\Theta' = 0$ becomes

$$\frac{4^2}{3}(a + b + c)^2 - \frac{4^2}{3}(a + b + c)^2 = 0,$$

which is satisfied identically. This is as it should be; for it is plain that there exists a triangle, viz. the triangle $(x = 0, y = 0, z = 0)$, inscribed in the conic $U = 0$, and circumscribed about the conic $V = 0$.

Suppose that the equation of the conic containing the angles is

$$y^2 - 4zx = 0,$$

and the equation of the conic touched by the sides is

$$ax^2 + by^2 + cz^2 = 0,$$

then the tangential equation of the last-mentioned conic is

$$bc\xi^2 + ca\eta^2 + ab\zeta^2 = 0;$$

and if we take for the angles of the triangle $x : y : z = 1 : 2\lambda : \lambda^2$, or $1 : 2\mu : \mu^2$, or $1 : 2\nu : \nu^2$, then the equation of the line joining the angles $(\mu), (\nu)$ is

$$2\mu\nu x - (\mu + \nu)y + z = 0,$$

which will touch the conic $ax^2 + by^2 + cz^2 = 0$ if

$$bc \cdot 4\mu^2\nu^2 + ca(\mu + \nu)^2 + ab \cdot 4 = 0;$$

and it is required to find under what circumstances the equations

$$bc \cdot 4\mu^2\nu^2 + ca(\mu + \nu)^2 + ab \cdot 4 = 0,$$

$$bc \cdot 4\nu^2\lambda^2 + ca(\nu + \lambda)^2 + ab \cdot 4 = 0,$$

$$bc \cdot 4\lambda^2\mu^2 + ca(\lambda + \mu)^2 + ab \cdot 4 = 0,$$

become equivalent to two equations only. The condition is of course included in the general formula; and putting

$$\text{disct.} \cdot (ka, kb + 1, kc, 0, -2, 0 \mid x, y, z)^2 = (K, \Theta, \Theta', K'_{\frac{1}{2}}k, 1)^3,$$

we must have

$$3\Theta^2 - 4K\Theta' = 0.$$

The discriminant in question is

$$k^3abc + k^2ac - k \cdot 4b - 4 = 0,$$

where $K = \frac{1}{2}ac$, $\Theta = -\frac{4}{3}b$, $K' = -4$; the required condition is therefore $ac + 16b^2 = 0$, or say

$$b = \pm \frac{1}{4}i\sqrt{ac}.$$

Substituting this value, the equations become

$$c\mu^2\nu^2 + i\sqrt{ca}(\mu + \nu)^2 + a = 0,$$

$$c\nu^2\lambda^2 + i\sqrt{ca}(\nu + \lambda)^2 + a = 0,$$

$$c\lambda^2\mu^2 + i\sqrt{ca}(\lambda + \mu)^2 + a = 0;$$

the first and second of these are

$$A + 2H + B\nu^2 = 0, \quad A' + 2H'\nu + B'\nu^2 = 0,$$

where

$$\begin{aligned} A &= (-i\sqrt{a} + \mu^2\sqrt{c})i\sqrt{a}, & H &= i\sqrt{ca}\mu, & B &= (i\sqrt{a} + \mu^2\sqrt{c})\sqrt{c}, \\ A' &= (-i\sqrt{a} + \lambda^2\sqrt{c})i\sqrt{a}, & H' &= i\sqrt{ca}\lambda, & B' &= (i\sqrt{a} + \lambda^2\sqrt{c})\sqrt{c}, \end{aligned}$$

and

$$\begin{aligned} AB' - 2HH' + A'B &= -2i\sqrt{ac}(a + c\lambda^2\mu^2) - 2i\sqrt{ac}\lambda\mu \cdot i\sqrt{ac}, \\ AB' - HH' &= -i\sqrt{ac}(a + i\sqrt{ac}\lambda\mu + c\mu^2), \\ H'A' - AH &= -i\sqrt{ac}(a - i\sqrt{ac}\lambda^2 + c\lambda^4); \end{aligned}$$

and the result of the elimination therefore is

$$(-a + i\sqrt{ac}\lambda^2 - c\lambda^4)(-a + i\sqrt{ac}\mu^2 - c\mu^4) + (-a + i\sqrt{ac}\lambda\mu - c\lambda^2\mu^2)^2 = 0,$$

viz.

$$2\sqrt{ca}(\lambda - \mu)^2(c\lambda^2\mu^2 + i\sqrt{ca}(\lambda + \mu)^2 + a) = 0;$$

which agrees, as it should do, with the third equation.

To find the condition that it may be possible in the conic

$$x^2 + y^2 + z^2 = 0$$

to inscribe an infinity of triangles, each of them circumscribed about the conic

$$ax^2 + by^2 + cz^2 = 0;$$

let the equations of the sides be

$$\begin{aligned} l\sqrt{a}x + m\sqrt{b}y + n\sqrt{c}z &= 0, \\ l'\sqrt{a}x + m'\sqrt{b}y + n'\sqrt{c}z &= 0, \\ l''\sqrt{a}x + m''\sqrt{b}y + n''\sqrt{c}z &= 0; \end{aligned}$$

then the conditions of circumscription are

$$\begin{aligned} l^2 + m^2 + n^2 &= 0, \\ l'^2 + m'^2 + n'^2 &= 0, \\ l''^2 + m''^2 + n''^2 &= 0; \end{aligned}$$

and the conditions of inscription are

$$\begin{aligned} bc(m'n'' - m''n')^2 + ca(n'l'' - n''l')^2 + ab(l'm'' - l''m')^2 &= 0, \\ bc(m''n - mn'')^2 + ca(n''l - nl'')^2 + ab(l''m - lm'')^2 &= 0, \\ bc(mn' - m'n)^2 + ca(nl' - n'l)^2 + ab(lm' - l'm)^2 &= 0. \end{aligned}$$

Now

$$\begin{aligned} (mn' - m'n)^2 &= (m^2 + n^2)(m'^2 + n'^2) - (mm' + nn')^2 \\ &= -l^2l'^2 - (mm' + nn')^2 \\ &= -mm' - mm' + (ll' + nn')(ll' + nn'); \end{aligned}$$

and making the like change in the analogous expressions, and putting for shortness

$$\begin{aligned} -bc + ca + ab &= \alpha, \\ bc - ca + ab &= \beta, \\ bc + ca - ab &= \gamma, \end{aligned}$$

the conditions in question become

$$\begin{aligned}(l'l'' + m'm'' + n'n'')(\alpha l'l'' + \beta m'm'' + \gamma n'n'') &= 0, \\ (l''l + m''m + n''n)(\alpha l''l + \beta m''m + \gamma n''n) &= 0, \\ (ll' + mm' + nn')(\alpha ll' + \beta mm' + \gamma nn') &= 0.\end{aligned}$$

The proper solution is that given by the system of equations

$$\begin{aligned}l^2 + m^2 + n^2 &= 0, \\ l'^2 + m'^2 + n'^2 &= 0, \\ l''^2 + m''^2 + n''^2 &= 0, \\ \alpha l'l'' + \beta m'm'' + \gamma n'n'' &= 0, \\ \alpha l''l + \beta m''m + \gamma n''n &= 0, \\ \alpha ll' + \beta mm' + \gamma nn' &= 0;\end{aligned}$$

and by writing $l = \frac{f}{\sqrt{\alpha}}$, $l' = \frac{f'}{\sqrt{\alpha}}$, $m = \frac{g}{\sqrt{\beta}}$, &c., $A = \frac{1}{\alpha}$, $B = \frac{1}{\beta}$, $C = \frac{1}{\gamma}$, these equations become

$$\begin{aligned}Af^2 + Bg^2 + Ch^2 &= 0, \\ Af'^2 + Bg'^2 + Ch'^2 &= 0, \\ Af''^2 + Bg''^2 + Ch''^2 &= 0, \\ f'f'' + g'g'' + h'h'' &= 0, \\ f''f + g''g + h''h &= 0, \\ ff' + gg' + hh' &= 0.\end{aligned}$$

The first of which systems expresses that the points (f, g, h) , (f', g', h') , (f'', g'', h'') are points in the conic

$$Ax^2 + By^2 + Cz^2 = 0;$$

and the second condition expresses that each of the points in question is the pole with respect to the conic

$$x^2 + y^2 + z^2 = 0$$

of the line joining the other two points, i.e. that the three points are a system of conjugate points with respect to the last-mentioned conic. The problem is thus reduced to the following one:—

To find the condition in order that it may be possible in the conic

$$Ax^2 + By^2 + Cz^2 = 0$$

to inscribe an infinity of triangles such that the angles are a system of conjugate points with respect to the conic

$$x^2 + y^2 + z^2 = 0.$$

Before going further it is proper to remark that if, instead of assuming

$$\alpha l'l'' + \beta m'm'' + \gamma n'n'' = 0,$$

we had assumed

$$l'l'' + m'm'' + n'n'' = 0,$$

this, combined with the equations

$$l'^2 + m'^2 + n'^2 = 0, \quad l''^2 + m''^2 + n''^2 = 0,$$

would have given $l' : m' : n' = l'' : m'' : n''$, i.e. two of the angles of the triangle would have been coincident: this obviously does not give rise to any proper solution. Returning now to the system of equations in f, g, h , &c., since the equations give only the ratios $f : g : h; f' : g' : h'; f'' : g'' : h''$, we may if we please assume

$$f^2 + g^2 + h^2 = 1, \quad f'^2 + g'^2 + h'^2 = 1, \quad f''^2 + g''^2 + h''^2 = 1,$$

which, combined with the second system of equations, gives

$$f + f' + f'' = 1, \quad g + g' + g'' = 1, \quad h + h' + h'' = 1.$$

We have, consequently,

$$\begin{aligned} A + B + C &= A(f^2 + f'^2 + f''^2) + B(g^2 + g'^2 + g''^2) + C(h^2 + h'^2 + h''^2) \\ &= (Af^2 + Bg^2 + Ch^2) + (Af'^2 + Bg'^2 + Ch'^2) + (Af''^2 + Bg''^2 + Ch''^2), \end{aligned}$$

i.e.

$$A + B + C = 0,$$

for the condition that it may be possible in the conic

$$Ax^2 + By^2 + Cz^2 = 0$$

to describe an infinity of triangles the angles of which are conjugate points with respect to the conic $x^2 + y^2 + z^2 = 0$.

The equation of the conic $Ax^2 + By^2 + Cz^2 = 0$ may be written in the form

$$(b^2c^2 - a^2bc + 2a^2bc)x^2 + (c^2a^2 - ab^2c + 2ab^2c)y^2 + (a^2b^2 - abc^2 + 2abc^2)z^2 = 0,$$

which gives the values of A, B, C ; or again in the form

$$\begin{aligned} &2(bc + ca + ab)(bcx^2 + cay^2 + abz^2) \\ &\quad - (bc + ca + ab)(x^2 + y^2 + z^2) \\ &\quad + 4abc(ax^2 + by^2 + cz^2) = 0; \end{aligned}$$

where it should be observed that $bcx^2 + cay^2 + abz^2 = 0$ is the equation of the conic which is the polar of $ax^2 + by^2 + cz^2 = 0$ with respect to $x^2 + y^2 + z^2 = 0$. It is very easy from the last form to deduce the equation of the auxiliary conic, when the conics $ax^2 + by^2 + cz^2 = 0$, $x^2 + y^2 + z^2 = 0$ are replaced by conics represented by perfectly general equations.

The condition $A + B + C = 0$ gives, substituting the values of A, B, C ,

$$b^2c^2 + a^2c^2 + a^2b^2 - 2abc(a + b + c) = 0;$$

or in a more convenient form,

$$(bc + ca + ab)^2 - 4abc(a + b + c) = 0,$$

as the condition in order that it may be possible to inscribe in the conic $x^2 + y^2 + z^2 = 0$ an infinity of triangles, the sides of which touch the conic $ax^2 + by^2 + cz^2 = 0$: this agrees perfectly with the general theorem.

It is convenient to add (as a somewhat more general form of the equation $A + B + C = 0$), that the condition in order that it may be possible in the conic $Ax^2 + By^2 + Cz^2 = 0$ to inscribe an infinity of triangles the angles of which are conjugate points with respect to the conic $A_1x^2 + B_1y^2 + C_1z^2 = 0$, is

$$\frac{A}{A_1} + \frac{B}{B_1} + \frac{C}{C_1} = 0.$$

But the problem to find the condition in order that it may be possible in the conic $x^2 + y^2 + z^2 = 0$ to inscribe an infinity of triangles the sides of which touch the conic $ax^2 + by^2 + cz^2 = 0$, may, by the assistance of the geometrical theorem to be presently mentioned, be at once reduced to the problem:—

To find the condition in order that it may be possible in the conic

$$x^2 + y^2 + z^2 = 0$$

to inscribe an infinity of triangles the sides of which are conjugate points with respect to a conic

$$A_1x^2 + B_1y^2 + C_1z^2 = 0.$$

The theorem referred to is as follows:—

Theorem. *If the chord PP' of a conic S envelope a conic σ , the points P, P' are harmonics with respect to a conic T which has with S, σ , a system of common conjugate points.*

Take for the equation of S ,

$$x^2 + y^2 + z^2 = 0,$$

and for the equation of σ ,

$$ax^2 + by^2 + cz^2 = 0;$$

then if $(x_1, y_1, z_1), (x_2, y_2, z_2)$ are the coordinates of the points P, P' respectively, we have

$$x_1^2 + y_1^2 + z_1^2 = 0, \quad x_2^2 + y_2^2 + z_2^2 = 0,$$

and the condition in order that the chord may touch the conic σ is

$$bc(y_1z_2 - z_1y_2)^2 + ca(z_1x_2 - x_1z_2)^2 + ab(x_1y_2 - y_1x_2)^2 = 0.$$

But we have

$$\begin{aligned} (y_1z_2 - y_2z_1)^2 &= (y_1^2 + z_1^2)(y_2^2 + z_2^2) - (y_1y_2 + z_1z_2)^2 \\ &= x_1^2x_2^2 - (y_1y_2 + z_1z_2)^2 \\ &= (x_1x_2 + y_1y_2 + z_1z_2)(x_1x_2 - y_1y_2 - z_1z_2); \end{aligned}$$

and making the like change in the analogous quantities, and putting for shortness

$$\begin{aligned} \alpha &= -bc + ca + ab, \\ \beta &= bc - ca + ab, \\ \gamma &= bc + ca - ab, \end{aligned}$$

the condition in question becomes

$$(x_1x_2 + y_1y_2 + z_1z_2)(\alpha x_1x_2 + \beta y_1y_2 + \gamma z_1z_2) = 0.$$

But the equation $x_1x_2 + y_1y_2 + z_1z_2 = 0$ must be rejected, as giving with the equations $x_1^2 + y_1^2 + z_1^2 = 0$, $x_2^2 + y_2^2 + z_2^2 = 0$ the relation $x_1 : y_1 : z_1 = x_2 : y_2 : z_2$; we have therefore

$$\alpha x_1x_2 + \beta y_1y_2 + \gamma z_1z_2 = 0,$$

which implies that the points (x_1, y_1, z_1) and (x_2, y_2, z_2) are harmonics with respect to the conic

$$\alpha x^2 + \beta y^2 + \gamma z^2 = 0,$$

which is a conic having with S, σ , a system of common conjugate points. The equation may also be written

$$(-bc + ca + ab)x^2 + (bc - ca + ab)y^2 + (bc + ca - ab)z^2 = 0;$$

or, as it may also be written,

$$(bc + ca + ab)(x^2 + y^2 + z^2) - 2(bc x^2 + ca y^2 + ab z^2) = 0;$$

and, as before remarked, $bc x^2 + ca y^2 + ab z^2 = 0$ is the equation of the conic which is the polar of $ax^2 + by^2 + cz^2 = 0$ with respect to $x^2 + y^2 + z^2 = 0$.

The condition in order that there may be inscribed in the conic $x^2 + y^2 + z^2 = 0$ an infinity of triangles the angles of which are conjugate points with respect to the conic $\alpha x^2 + \beta y^2 + \gamma z^2 = 0$, is

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = 0;$$

or writing this equation under the form $\beta\gamma + \gamma\alpha + \alpha\beta = 0$, and substituting for α, β, γ their values, we have the equation already found, as the condition in order that it may be possible in the conic $x^2 + y^2 + z^2 = 0$ to inscribe an infinity of triangles the sides of which touch the conic $ax^2 + by^2 + cz^2 = 0$.

Theorem. *Let*

$$ax^2 + by^2 + cz^2 = 0$$

be the equation of a spherical conic, and let $(\xi : \eta : \zeta)$, a point on the conic, be the pole of a great circle cutting the conic in two points; the conic intersects upon the great circle an arc given by the equation

$$\cos \delta = \frac{(a + b + c)\sqrt{\xi^2 + \eta^2 + \zeta^2}}{\sqrt{(a + b + c)^2(\xi^2 + \eta^2 + \zeta^2) - 4(bc\xi^2 + ca\eta^2 + ab\zeta^2)}};$$

hence if $a + b + c = 0$, $\delta = 90^\circ$; or there may be inscribed in the conic an infinity of triangles having each of their sides equal to 90° .

It is worth while, in connexion with the subject, and for the sake of a remark to which they give rise, to reproduce in a short compass some results long ago obtained by Jacobi

and Richelot. The following are Jacobi's formulae for the chords of a circle subjected to the condition of touching another circle; viz. if in the figure we put

$$\begin{aligned} CP &= R, \\ cp &= r, \\ Cc &= a, \\ \angle ACP &= 2\phi, \\ \angle ACP' &= 2\phi', \end{aligned}$$

then it is clear from geometrical considerations that

$$\frac{d\phi}{MA} = \frac{d\phi'}{MA'}.$$

We have

$$\begin{aligned} \overline{MA}^2 &= \overline{cA}^2 - \overline{cM}^2 = a^2 + R^2 + 2aR \cos 2\phi - r^2 \\ &= (a + R)^2 - r^2 - 4aR \sin^2 \phi \\ &= [(a + R)^2 - r^2](1 - k^2 \sin^2 2\phi), \end{aligned}$$

or

$$\frac{d\phi}{\sqrt{1 - k^2 \sin^2 \phi}} = \frac{d\phi'}{\sqrt{1 - k^2 \sin^2 \phi'}},$$

where

$$k^2 = \frac{4aR}{(a + R)^2 - r^2},$$

and therefore also

$$k'^2 = \frac{(R - a)^2 - r^2}{(R + a)^2 - r^2}.$$

It will be convenient for comparison with the formulae of Richelot to write $\angle ACQ = 2\psi$; this gives $2\psi = \pi - 2\phi$, and the differential equation thus becomes

$$\frac{d\psi}{\sqrt{a^2 + R^2 - r^2 - 2aR \cos 2\psi}} = \frac{d\psi'}{\sqrt{a^2 + R^2 - r^2 - 2aR \cos 2\psi'}},$$

i.e.

$$\frac{d\psi}{\sqrt{m - n \cos 2\psi}} = \frac{d\psi'}{\sqrt{m - n \cos 2\psi'}},$$

or if

$$\tan \psi = \frac{m-n}{m+n} \tan \theta,$$

and therefore

$$\begin{aligned} \cos 2\psi &= \frac{1 - \frac{(m-n)^2}{(m+n)^2} \tan^2 \theta}{1 + \frac{(m-n)^2}{(m+n)^2} \tan^2 \theta}, \\ m - n \cos 2\psi &= \frac{(m - n) \cos^2 \theta + (m + n)^{-1} (m - n)^2 \sin^2 \theta}{\cos^2 \theta + \frac{(m-n)^2}{(m+n)^2} \sin^2 \theta}, \end{aligned}$$

if $g^2 = \frac{m-n}{m+n}$, then this is

$$= \frac{m-n}{\cos^2 \theta + \frac{m-n}{m+n} \sin^2 \theta},$$

and we have also

$$d\psi = \frac{\sqrt{\frac{m-n}{m+n}} d\theta}{1 - \frac{2n}{m+n} \sin^2 \theta},$$

and thence

$$\frac{d\psi}{\sqrt{m-n \cos 2\psi}} = \frac{d\theta}{\sqrt{m+n} \sqrt{1 - \frac{2n}{m+n} \sin^2 \theta}},$$

that is,

$$\frac{d\psi}{\sqrt{m-n \cos 2\psi}} = \frac{d\theta}{\sqrt{m+n} \sqrt{1 - k^2 \sin^2 \theta}},$$

where $k^2 = \frac{2n}{m+n}$ has the same value as before. Hence the relation between θ, θ' is

$$\frac{d\theta}{\sqrt{1 - k^2 \sin^2 \theta}} = \frac{d\theta'}{\sqrt{1 - k^2 \sin^2 \theta'}},$$

which is identical with that between ϕ and ϕ' ; and in fact the equation between θ, ϕ is

$$\tan \theta \tan \phi = g',$$

which, if $\phi = \text{am } u$, gives $\theta = \text{am}(K' - u)$.

The differential equation contains only a single arbitrary parameter; hence the same differential equation might have been obtained from different values of a, R, r , the parameters which determine the circle enveloped by the moveable chord. The condition for this of course is $\frac{m'}{n'} = \frac{m}{n}$, that is

$$\frac{a^2 + R^2 - r^2}{2aR} = \frac{a'^2 + R'^2 - r'^2}{2a'R'},$$

or as the equation may be written

$$(aa' - RR')(a - a') - R(a(r'^2 - ar'^2)) = 0;$$

this implies that the enveloped circles intersect the other circle in the same two points, or that all the circles have a common chord.

Suppose for $\psi = 0$, we have $\psi' = \beta$, then it is easy to see geometrically that

$$\tan^2 \beta = \frac{(R-a)^2 - r^2}{r^2};$$

let the corresponding value of θ' be $\theta' = \alpha$, i.e. suppose that for $\theta = 0$, we have $\theta' = \alpha$, then

$$\tan^2 \alpha = \frac{(R+a)^2 - r^2}{(R-a)^2 - r^2} \cdot \frac{(R-a)^2 - r^2}{r^2},$$

i.e.

$$\tan^2 \alpha = \frac{(R+a)^2 - r^2}{r^2};$$

or, what is the same thing,

$$\sec \alpha = \frac{R}{r} + \frac{a}{r};$$

and α having this value, we have for the finite relation between θ, θ' ,

$$F\theta' = F\theta + F\alpha.$$

Richelot has shown, by precisely similar reasoning, that for circles of the sphere we have

$$\frac{d\psi}{\sqrt{\cos^2 r - (\cos R \cos a + \sin R \sin a \cos 2\psi)^2}} = \frac{d\psi'}{\sqrt{\cos^2 r - (\cos R \cos a + \sin R \sin a \cos 2\psi')^2}},$$

which is of the form

$$\frac{d\psi}{\sqrt{1 - (\lambda + \mu \cos 2\psi)^2}} = \frac{d\psi'}{\sqrt{1 - (\lambda + \mu \cos 2\psi')^2}},$$

where

$$\lambda = \frac{\cos R \cos a}{\cos r},$$

$$\mu = \frac{\sin R \sin a}{\cos r}.$$

It is very important to remark, that this equation contains the two parameters λ, μ , so that the same equation cannot be obtained with any new values of the parameters a, r ; or the formulae in *plano* for three or more circles do not apply to circles of the sphere: the geometrical reason for this is as follows, viz. in the plane a circle is a conic passing through two fixed points (the circular points at ∞), and consequently any number of circles having a common chord are in fact to be considered as conics each of which passes through the same four points. But circles of the sphere are not spherical conics passing through two fixed points, but are merely spherical conics having a double contact with an imaginary spherical conic (viz. the curve of intersection of the sphere with a sphere radius zero); hence circles of the sphere having a common spherical chord are not spherical conics passing through the same four points. I am not sure whether this remark as to the ground of the distinction between the theory of circles in *plano* and that of circles on the sphere has been explicitly made in any of the treatises on spherical geometry.

To reduce the equation, write

$$\tan \psi = \sqrt{\frac{1 - (\lambda + \mu)}{1 - (\lambda - \mu)}} \tan \theta,$$

then, after a simple reduction,

$$\frac{d\psi}{\sqrt{1 - (\lambda + \mu \cos 2\psi)^2}} = \frac{d\theta}{\sqrt{(1 + \mu)^2 - \lambda^2} \sqrt{1 - \frac{4\mu}{(1 + \mu)^2 - \lambda^2} \sin^2 \theta}},$$

or the relation between the two values of θ is

$$\frac{d\theta}{\sqrt{1 - k^2 \sin^2 \theta}} = \frac{d\theta'}{\sqrt{1 - k^2 \sin^2 \theta'}},$$

where

$$k^2 = \frac{4\mu}{(1 + \mu)^2 - \lambda^2},$$

that is

$$k^2 = \frac{4 \frac{\tan R}{\tan r} \cdot \frac{\sin a}{\cos R \sin r}}{\left(\frac{\tan R}{\tan r} + \frac{\sin a}{\cos R \sin r} \right)^2 - 1}.$$

Suppose that for $\psi = 0$, we have $\psi' = \beta$, it is easy to see that

$$\tan^2 \beta = \frac{\sin^2(R - a) - \sin^2 r}{\cos^2 r};$$

let the corresponding value of θ' be $\theta' = \alpha$, i.e. suppose that for $\theta = 0$, we have $\theta' = \alpha$, then

$$\begin{aligned} \tan^2 \alpha &= \frac{1 - \frac{\cos(R + a)}{\cos r}}{1 - \frac{\cos(R - a)}{\cos r}} \cdot \frac{\sin^2(R - a) - \sin^2 r}{\cos^2 r} \\ &= \frac{\cos r - \cos(R + a)}{\cos r - \cos(R - a)} \cdot \frac{\cos^2 r - \cos^2(R - a)}{\cos^2 r} \\ &= \frac{(\cos r - \cos(R + a))(\cos r + \cos(R - a))}{\cos^2 r} \\ &= \frac{\sin R \sin a + \cos r \cos a - \cos^2 R \cos^2 a}{\cos^2 R \sin^2 r} \\ &= \frac{(\cos r + \sin R \sin a)^2 - \cos^2 R \cos^2 a}{\cos^2 R \sin^2 r}, \end{aligned}$$

i.e.

$$\tan^2 \alpha = \left(\frac{\tan R}{\tan r} + \frac{\sin a}{\cos R \sin r} \right)^2 - 1,$$

whence

$$\sec \alpha = \left(\frac{\tan R}{\tan r} + \frac{\sin a}{\cos R \sin r} \right);$$

and α having this value, the finite relation between θ, θ' is

$$F\theta' = F\theta + F\alpha.$$

By comparing with the corresponding formula in *plano*, we arrive at Richelot's conclusion, that the formulae for the sphere may be deduced from those in *plano* by writing in the place of $\frac{R}{r}, \frac{a}{r}$, the functions $\frac{\tan R}{\tan r}, \frac{\sin a}{\cos R \sin r}$, respectively.

2, Stone Buildings, October 1, 1856.

203. On the Theory of the Analytical Forms called Trees

[From the *Philosophical Magazine*, vol. XIII. (1857), pp. 172–176.]

A symbol such as $A dx + B dy + \dots$, where $A, B, \&c.$ contain the variables $x, y, \&c.$ in respect to which the differentiations are to be performed, partakes of the natures of an operand and operator, and may be therefore called an Operandator. Let $P, Q, R, \dots$ be any operandators, and let U be a symbol of the same kind, or to fix the ideas, a mere operand; PU denotes the result of the operation P performed on U , and QPU denotes the result of the operation Q performed on PU ; and generally in such combinations of symbols, each operation is considered as affecting the operand denoted by means of all the symbols on the right of the operation in question. Now considering the expression QPU , it is easy to see that we may write

$$QPU = (Q \times P)U + (QP)U,$$

where on the right-hand side $(Q \times P)$ and (QP) signify as follows: viz. $Q \times P$ denotes the mere algebraical product of Q and P , while QP (consistently with the general notation as before explained) denotes the result of the operation Q performed upon P as operand; and the two parts $(Q \times P)U$ and $(QP)U$ denote respectively the results of the operations $(Q \times P)$ and (QP) performed each of them upon U as operand. It is proper to remark that $(Q \times P)$ and $(P \times Q)$ have precisely the same meaning, and the symbol may be written in either form indifferently. But without a more convenient notation, it would be difficult to find the corresponding expressions for $RQPU$, &c. This, however, can be at once effected by means of the analytical forms called trees (see figs. 1, 2, 3), which contain all the trees which can be formed with one branch, two branches, and three branches respectively.

The inspection of these figures will at once show what is meant by the term in question, and by the terms *root*, *branches* (which may be either main branches, intermediate branches, or free branches), and *knots* (which may be either the root itself, or proper knots, or the extremities of the free branches). To apply this to the question in hand, PU consists of a single term represented by fig. 1 (*bis*); QPU consists, as above, of two terms represented by the two parts of fig. 2 (*bis*), viz. the first part represents the term $(Q \times P)U$, and the second part represents the term $(QP)U$. And it is obvious that fig. 2 (*bis*) is at once formed from the figure 1 (*bis*) by adding on a branch terminated by Q at each of the knots of the single part of fig. 1 (*bis*). In like manner $RQPU$ consists of six terms represented by the six parts of fig. 3 (*bis*), and this figure is at once formed from fig. 2 (*bis*) by adding on a branch terminated by R at each knot of each part of fig. 2 (*bis*). It is hardly necessary to remark that the first part of fig. 3 (*bis*) denotes what, in the notation first explained, would be denoted by $(R \times Q \times P)U$, the second term what would in like manner be denoted by $(RQ \times P)U$, and so on; the last part being the term which would be denoted by $((RQ)P)U$, viz. R operates upon Q , giving the operandator RQ , which operates upon P , giving the operandator $(RQ)P$, which finally operates upon U .

The figures 1 (*bis*), 2 (*bis*), &c. contain the same trees as are contained in the corresponding figures 1, 2, &c.; only, on account of the different modes of filling up, trees are considered as so many distinct trees in a figure of the second set which are considered as one and the same tree in the corresponding figure of the first set. A difference in the number of trees first occurs in the figures 3 and 3 (*bis*), the first of which contains only

four, while the latter contains six trees, viz. the first tree, the second, third and fourth trees, the fifth tree and the sixth tree of fig. 3 (*bis*) correspond respectively to the first tree, the second tree, the third tree, and the fourth tree of fig. 3. To derive fig. 3 (*bis*) from fig. 3, we must fill up the trees of fig. 3 with U at the root and R, Q, P at the other knots in every possible manner, subject only to the restriction, that, reckoning up from the extremity of a free branch to the root, there must not be any transposition in the order of the symbols RQP , and taking care to admit only distinct trees. Thus the first tree of fig. 3 might be filled up in six ways; but the trees so obtained are considered as one and the same tree, and we have only the first tree of fig. 3 (*bis*). Again, on account of the restriction, the fourth tree of fig. 3 can be filled up in one way only, and we have thus the sixth tree of fig. 3 (*bis*). And thus, in general, each figure of the second set can be formed at once from the corresponding figure of the first set; or when the first set of figures is given, the expression for $YX \dots QPU$ can be formed directly without the assistance of the expression for the preceding symbol $X \dots QPU$; the number of terms for the n th figure of the second set is obviously $1.2.3 \dots n$, and consequently it is only necessary to count the terms in order to ascertain that no admissible mode of filling up has been omitted.

The number of parts in any one of the figures of the first set is much smaller than the number of parts in the corresponding figure of the second set; and the law for the number of parts, i.e. for the number n of the trees with n branches, is a very singular one. To obtain this law, we must consider how the trees with n branches can be formed by means of those of a smaller number of branches. A tree with n branches has either a single main branch, or else two main branches, three main branches, &c.,... to n main branches. If the tree has one main branch, it can only be formed by adding on to this main branch a tree with $(n - 1)$ branches, i.e. A_n contains a part A_{n-1} . If the tree has two main branches, then $p + q$ being a partition of $n - 2$, the tree can be formed by adding on to one main branch a tree of p branches, and to the other main branch a tree of q branches; the number of trees so obtained is $A_p A_q$; this, however, assumes that the parts p and q are unequal; if they are equal, it is easy to see that the number of trees is only $\frac{1}{2}A_p(A_p + 1)$. Hence $p + q$ being any partition of $n - 2$, A_n contains the part $A_p A_q$ if p and q are unequal, and the part $\frac{1}{2}A_p(A_p + 1)$ if p and q are equal. In like manner, considering the trees with three main branches, then if $p + q + r$ is any partition of $n - 3$, A_n contains the part $A_p A_q A_r$ if p, q, r are unequal; but if two of these numbers, e.g. p and q , are equal, then the part $\frac{1}{2}A_p(A_p + 1) A_r$; and if p, q, r are all equal, then the part $\frac{1}{6}A_p(A_p + 1)(A_p + 2)$; and so on, until lastly we have a single tree with n main branches, or A_n contains the part unity. A little consideration will show that the preceding rule for the formation of the number A_n is completely expressed by the equation

$$(1 - x)^{-A_1}(1 - x^2)^{-A_2}(1 - x^3)^{-A_3} \dots = 1 + A_1x + A_2x^2 + A_3x^3 + \&c.,$$

and consequently that we may, by means of this equation, calculate successively for the different values of n the number A_n of the trees with n branches. The calculation may be effected very easily as follows: [the table as originally printed contained at the end of it some errors of calculation which were corrected, B.A. Report for 1875, p. 258].

$A_1 =$	1	(1)	1	1	1	1	1	1	1	1	1	1	1	1	1
			1	1	1	1	1	1	1	1	1	1	1	1	1
					1	1	1	1	1	1	1	1	1	1	1
							1	1	1	1	1	1	1	1	1
								1	1	1	1	1	1	1	1
$A_2 =$	1	1	(2)	2	3	3	4	4	5	5	6	6	7	7	1
				2	2	4	4	6	6	8	8	10	10	12	1
						3	3	6	6	9	9	12	12	15	1
								4	4	8	8	12	12	16	1
$A_3 =$	1	1	2	(4)	3	7	11	13	17	23	27	38	47	61	7
				1	4	8	16	20	28	38	50	68	88	116	14
						4	8	16	20	28	40	50	68	88	11
								10	10	20	30	40	60	80	11
$A_4 =$	1	1	2	4	(9)	11	19	29	47	66	91	109	167	231	29
					9	9	18	27	45	54	81	99	162	207	29
$A_5 =$	1	1	2	4	9	(20)	26	47	83	142	235	357	561	873	132
						20	20	40	60	100	140	200	280	400	56
$A_6 =$	1	1	2	4	9	20	(48)	67	123	222	415	752	1284	2228	374
							48	48	96	144	240	336	528	720	105
$A_7 =$	1	1	2	4	9	20	48	(115)	171	318	607	1144	2104	3838	692
								115	115	230					
$A_8 =$	1	1	2	4	9	20	48	115	(286)	433	837	1626	3138	5910	1099
									286	286					
$A_9 =$	1	1	2	4	9	20	48	115	286	(719)	1123	2230	4424	8716	1696
										719					
$A_n =$	1	2	4	9	20	48	115	286	719	1842					
for $r =$	1	2	3	4	5	6	7	8	9	10					

I have had occasion, for another purpose, to consider the question of finding the number of trees with a given number of free branches, bifurcations at least. Thus, when the number of free branches is three, the trees of the form in question are those in the annexed figure, and the number is therefore two. It is not difficult to see that we have in this case (B_r being the number of such trees with r free branches),

$$(1-x)^{-B_1}(1-x^2)^{-B_2}(1-x^3)^{-B_3}\dots = 1+x+2B_1x^3+2B_2x^4+2B_3x^5+\&c.;$$

and a like process of development gives:

$B_r =$	1	2	5	12	33	90
for $r =$	2	3	4	5	6	7

I may mention, in conclusion, that I was led to the consideration of the foregoing theory of trees by Professor Sylvester's researches on the change of the independent variables in the differential calculus.

2, Stone Buildings, January 2, 1856.

204. On a Problem in the Partition of Numbers

[From the *Philosophical Magazine*, vol. XIII. (1857), pp. 245–248.]

It is required to find the number of partitions into a given number of parts, such that the first part is unity, and that no part is greater than twice the preceding part.

Commencing to form the partitions in question, these are

$$\begin{array}{l} 1 \mid 1 \mid 1 \mid 1 \mid 1 \mid 1 \mid 1 \mid 1 \mid 1 \text{ \&c.}; \\ 2 \mid 1 \mid 1 \mid 2 \mid 2 \mid 2 \\ 1 \mid 2 \mid 1 \mid 2 \mid 3 \mid 4 \end{array}$$

and if we were to proceed to the 4-partitions, each 3-partition ending in 1 would give rise to two such partitions; each 3-partition ending in 2 to four such partitions; each 3-partition ending in 3 to six partitions; and each 3-partition ending in 4 to eight such partitions. We form in this manner the Table:

Number of	ending in																Totals
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
1-partitions	1																1
2-partitions	1	1															2
3-partitions	2	2	1	1													6
4-partitions	6	6	4	4	2	2	1	1									26
5-partitions &c.	26	26	20	20	14	14	10	10	6	6	4	4	2	2	1	1	166

and we are thus led to the series

$$\begin{array}{l} 1, \\ 1, 2, \\ 1, 2, 4, 6, \\ 1, 2, 4, 6, 10, 14, 20, 26, \\ \text{\&c.}; \end{array}$$

where, considering 0 as the first term of each series, the first differences of any series are the terms twice repeated of the next preceding series: thus the differences of the fourth series are 1, 1, 2, 2, 4, 4, 6, 6. It is moreover clear that the first half of each series is precisely the series which immediately precedes it; we need, in fact, only consider a single infinite series, 1, 2, 4, 6, &c. It is to be remarked, moreover, that in the column of totals, the total of any line is precisely the first number in the next succeeding line.

Consider in general a series $A, B, C, D, E, \text{\&c.}$, and a series $A', B', C', D', E', \text{\&c.}$ de-

rived from it as follows:

$$\begin{aligned}
 A' &= 1A, \\
 B' &= 2A, \\
 C' &= 2A + B, \\
 D' &= 2A + 2B, \\
 E' &= 2A + 2B + C, \\
 F' &= 2A + 2B + 2C, \\
 &\&c.;
 \end{aligned}$$

viz. the first differences of the series $0, A', B', C', D', E', \&c.$ are $A, A, B, B, C, C, \&c.$ Then multiplying by $1, x, x^2, \&c.$ and adding, we have

$$\begin{aligned}
 A' + B'x + C'x^2 + \&c. &= (1 + 2x + 2x^2 + \cdots)(A + Bx^2 + Cx^4 + \&c.) \\
 &= \frac{1+x}{1-x}(A + Bx^2 + Cx^4 + \&c.);
 \end{aligned}$$

and if we form in a similar manner from $A', B', C', D', \&c.$ a series $A'', B'', C'', D'', \&c.$ and so on, we have

$$\begin{aligned}
 A'' + B''x + C''x^2 + \&c. &= \frac{1+x}{1-x}(A' + B'x^2 + C'x^4 + \&c.) \\
 &= \frac{1+x}{1-x} \cdot \frac{1+x^2}{1-x^2}(A + Bx^4 + Cx^8 + \&c.),
 \end{aligned}$$

and so on. Write $A = 1$, and suppose that the process is repeated an indefinite number of times, we have

$$1 + \mathfrak{B}x + \mathfrak{C}x^2 + \mathfrak{D}x^3 + \&c. = \frac{1+x}{1-x} \cdot \frac{1+x^2}{1-x^2} \cdot \frac{1+x^4}{1-x^4} \cdot \&c.,$$

and the coefficients $1, \mathfrak{B}, \mathfrak{C}, \mathfrak{D}, \&c.$ are precisely those of the infinite series $1, 2, 4, 6, \&c.$ We have more simply

$$1 + \mathfrak{B}x + \mathfrak{C}x^2 + \mathfrak{D}x^3 + \&c. = \frac{1}{(1-x)^2(1-x^2)(1-x^4)(1-x^8)\&c.},$$

which gives rise to the following very simple algorithm for the calculation of the coefficients:

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
0	0	1	2	4	6	9	12	16	20	25	30	36	42	49	56
1	2	4	6	9	12	16	20	25	30	36	42	49	56	64	72
0	0	0	0	1	2	4	6	10	14	20	26	35	44	56	68
1	2	4	6	10	14	20	26	35	44	56	68	84	100	120	140
0	0	0	0	0	0	0	0	1	2	4	6	10	14	20	26
1	2	4	6	10	14	20	26	36	46	60	74	94	114	140	166
$\&c.$															

The last line is marked off into periods of (reckoning from the beginning) 1, 2, 4, 8, &c.; and by what has preceded, the series which gives the number of 1-partitions, 2-partitions, 3-partitions, &c. is found by summing to the end of each period and doubling the results; we thus, in fact, obtain (1), 2, 6, 26, 166, 1626, &c.: and the same series is also given by means of the last terms of the several periods.

The preceding expression for $1 + \mathfrak{B}x + \mathfrak{C}x^2 + \&c.$ shows that $\mathfrak{B}, \mathfrak{C}, \&c.$ are the number of partitions of 1, 2, 3, 4, 5, 6, &c. respectively into the parts 1, 1', 2, 4, 8, &c.; and we are thus led to

Theorem. *The number of λ -partitions (first part unity, no part greater than twice the preceding one) is equal to the number of partitions of $2^{\lambda-1} - 1$ into the parts 1, 1', 2, 4, $\dots 2^{\lambda-2}$. Or, again, it is equal to twice the sum of the number of partitions of $0, 1, 2, \dots 2^{\lambda-1} - 1$ respectively into the parts 1, 1', 2, 4, $\dots 2^{\lambda-3}$ (where the number of partitions of 0 counts for 1).*

For example, the partitions of 0, 1, 2, 3, &c. with the parts 1, 1', 2, $\dots$ are

$$\begin{aligned} &() \\ &1, 1' \\ &1 + 1, 1 + 1', 1' + 1', 2 \\ &1 + 1 + 1, 1 + 1 + 1', 1 + 1' + 1', 1' + 1' + 1', 2 + 1, 2 + 1', \end{aligned}$$

the numbers of which are 1, 2, 4, 6. Hence, by the first part of the theorem, the number of 3-partitions is 6, and by the second part of the theorem, the number of 4-partitions is

$$2(1 + 2 + 4 + 6) = 26.$$

2, Stone Buildings, March 17, 1857.

205. Note on the Summation of a Certain Factorial Expression

[From the *Philosophical Magazine*, vol. XIII. (1857), pp. 419–423.]

Mr Kirkman some months ago communicated to me a formula for the double summation of a factorial expression, to which formula he had been led by his researches on the partition of polygons. The formula in a slightly altered form is as follows: viz.

$$\sum_x \sum_y \frac{[x+y+2]^y}{[y+1]^{y+1}} \cdot \frac{[x]^x}{[y]^y} \cdot \frac{[r+k-x-y]^{r-1-x-y}}{[k-y]^{k-y}} \cdot \frac{[r-1-x]^{x-1-x}}{[k-1-y]^{k-1-y}} = \frac{2k}{r+3} \cdot \frac{[r+k+2]^k [r]^r}{[k+1]^{k+1} [k]^k},$$

the summation extending from $x = 0$ to $x = r - 1$, and $y = 0$ to $y = k - 1$. In the particular case when $k = r$, then all the terms of the series except those in which $y = x$ vanish; and putting therefore $k = r$ and $y = x$, and making a slight change in the form of the right-hand side, the formula becomes

$$\sum \frac{[2x+2]^x}{[x+1]^{x+1}} \cdot \frac{(2r-2x)^{r-1-x}}{[r-x]^{r-x}} = 4 \frac{[2r+1]^{r-1}}{[r-1]^{r-1}},$$

the summation extending from $x = 0$ to $x = r - 1$.

We have, in the notation of Gauss, $[m]^n = m(m-1) \cdots 2 \cdot 1 = \Pi m$, and a factorial $[m]^n$ is expressed in terms of the function Π by the formula $[m]^n = \Pi m \div \Pi(m-n)$. Write also

$$\Pi_1(m - \tfrac{1}{2}) = (m - \tfrac{1}{2})(m - \tfrac{3}{2}) \cdots \tfrac{3}{2} \cdot \tfrac{1}{2},$$

we have

$$\begin{aligned} \Pi(2m) &= 2^{2m} \Pi m \Pi_1(m - \tfrac{1}{2}), \\ \Pi(2m+1) &= 2^{2m+1} \Pi m \Pi_1(m + \tfrac{1}{2}); \end{aligned}$$

and transforming the factorials by these formulae, the series becomes

$$\sum \frac{\Pi_1(x + \tfrac{1}{2})}{\Pi_1(\tfrac{1}{2})} \cdot \frac{\Pi_1(r-x-\tfrac{1}{2})}{\Pi_1(r-\tfrac{1}{2})} \cdot \frac{\Pi 2}{\Pi(x+2) \Pi(r-x+1)} = \frac{8r(r+\tfrac{1}{2})}{(r+2)(r+3)} \cdot \frac{\Pi(r+1)}{\Pi(r+3)},$$

the summation, as before, from $x = 0$ to $x = r - 1$. This may be written

$$\sum \frac{4(r+\tfrac{1}{2})}{r+2} \cdot \frac{-8(r+\tfrac{1}{2})(r+\tfrac{1}{2})}{(r+2)(r+3)} + \frac{\Pi 2}{(x-2)} \cdot \frac{\Pi(r+1)}{(r-2)(r-3)} = \frac{8r(r+\tfrac{1}{2})}{(r+2)(r+3)},$$

the summation from $x = 0$ to $x = r - 1$. The general term does not vanish for $x = r$ or $x = r + 1$, but it vanishes for all greater values of x ; hence if we add to the right-hand side the two terms corresponding to $x = r$ and $x = r + 1$, the summation may be extended from $x = 0$ to $x = r + 1$, or what is the same thing, from $x = 0$ indefinitely. The two terms in question are

$$\frac{r+2}{r+2} + \frac{(r+2)(r+3)}{(r+2)(r+3)} - \frac{4r(r+\tfrac{1}{2})}{(r+2)(r+3)},$$

and the resulting equation is

$$\sum \frac{\Pi_1(x + \tfrac{1}{2}) \Pi_1(r-x-\tfrac{1}{2}) \Pi 2}{\Pi_1(\tfrac{1}{2}) \Pi_1(r-\tfrac{1}{2}) \Pi(x+2) \Pi(r-x+1)} = \frac{4r(r+\tfrac{1}{2})}{\Pi(r+1)},$$

the summation from $x = 0$ indefinitely; or substituting for the functions Π and Π_1 their values, the formula is

$$\frac{1}{3 \cdot (r + \frac{3}{2})} - \frac{3 \cdot (r - \frac{1}{2})}{3 \cdot 4 \cdot (r + \frac{3}{2})(r + \frac{5}{2})} - \frac{3 \cdot 4 \cdot 5 \cdot (r - \frac{1}{2})(r - \frac{3}{2})}{3 \cdot 4 \cdot 5 \cdot (r + \frac{3}{2})(r + \frac{5}{2})(r + \frac{7}{2})} + \dots = \frac{4r(r + \frac{1}{2})}{(r + 2)(r + 3)},$$

which is a formula obviously belonging to the theory of the hypergeometric series

$$F(\alpha, \beta, \gamma, x) = 1 + \frac{\alpha \cdot \beta}{1 \cdot \gamma}x + \frac{\alpha \cdot \alpha + 1 \cdot \beta \cdot \beta + 1}{1 \cdot 2 \cdot \gamma \cdot \gamma + 1}x^2 + \&c.;$$

but the formula applicable to the case in hand has probably not been given. It may be proved as follows, premising that I disregard all difficulties arising from infinite values of the functions in the definite integrals, convergency, &c. We have

$$\int_0^1 \theta^{\alpha-1} (1-\theta)^{-\gamma-\alpha+\beta-1} (1-x\theta)^{-\beta} d\theta = \frac{\Pi(\alpha-1) \Pi(-\gamma-\alpha+\beta-1)}{\Pi(-\gamma-1)} \left(1 + \frac{\alpha \cdot \beta}{1 \cdot \gamma}x + \frac{\alpha \cdot \alpha + 1 \cdot \beta \cdot \beta + 1}{1 \cdot 2 \cdot \gamma \cdot \gamma + 1}x^2 + \dots \right)$$

Now we have

$$\int_0^1 dx \int_0^1 d\theta (1-x\theta)^{-\beta} = \frac{1}{(\beta+1)(\beta+2)\theta^2} (1-x\theta)^{-\beta+2} - \frac{1}{(\beta+1)(\beta+2)} \cdot \frac{1}{\theta^2} + \frac{x}{\theta(\beta+1)},$$

and hence multiplying by $d\alpha$ and integrating from $x = 0$, and again multiplying by $d\alpha$ and integrating from $x = 0$ to $x = 1$, we find

$$\int_0^1 d\theta \theta^{\alpha-3} (1-\theta)^{-\gamma-\alpha+\beta-1} = \int_0^1 d\theta \theta^{\alpha-1} (1-\theta)^{-\gamma-\alpha+\beta-1} + (\beta+2) \int_0^1 d\theta \theta^{\alpha-1} (1-\theta)^{-\gamma-\alpha+\beta-1} \cdot \frac{1}{2} S,$$

if for shortness

$$S = 1 + \frac{\alpha \cdot \beta}{3 \cdot \gamma} + \frac{\alpha \cdot \alpha + 1 \cdot \beta \cdot \beta + 1}{3 \cdot 4 \cdot \gamma \cdot \gamma + 1} + \&c.$$

Substituting for the definite integrals their values,

$$\frac{\Pi(\alpha-3) \Pi(-\gamma+\beta-1)}{\Pi(-\gamma+\beta-3)} = \frac{\Pi(\alpha-1) \Pi(-\gamma-\alpha-2)}{\Pi(-\gamma-1)} + (\beta+2) \frac{\Pi(\alpha-1) \Pi(-\gamma-\alpha-2)}{\Pi(-\gamma-1)} \cdot \frac{1}{2} S,$$

whence

$$\frac{1}{2}(\beta+1)(\beta+2)S = \frac{\Pi(\alpha-3) \Pi(-\alpha-\gamma+\beta+1)}{\Pi(\alpha-1) \Pi(-\alpha-\gamma-1)} \cdot \frac{\Pi(-\gamma-1)}{\Pi(-\gamma+\beta-1)} - \frac{\Pi(\alpha-3) \Pi(-\gamma-1)}{\Pi(\alpha-1) \Pi(-\gamma-2)} + (\beta+2) \frac{\Pi(\alpha-3)}{\Pi(\alpha-1)}.$$

The second and third terms are

$$-\frac{1}{(\alpha-1)(\alpha-2)}(\gamma+1)(\gamma+2) - (\beta+2) \frac{1}{\alpha-1}(\gamma+1),$$

which are

$$= -\frac{\gamma+1}{(\alpha-1)(\alpha-2)}(\alpha\beta+2\alpha+2\beta+\gamma-2).$$

For the reduction of the first term we have

$$\Pi(-\alpha-\gamma+\beta-1) = [\beta-\gamma-1]^\beta \Pi(-\alpha-\gamma-1);$$

$$\Pi(\beta - \gamma - 1) = [\beta - \gamma - 1]^\beta \Pi(-\gamma - 1);$$

and we thus find

$$\frac{1}{2}(\beta+1)(\beta+2)(\alpha-1)(\alpha-2)S = \frac{[\beta - \gamma - 1]^\beta \Pi(-\alpha - \gamma + \beta - 1) \Pi(-\gamma - 1)}{[\beta - \gamma - 1]^\beta \Pi(-\alpha - \gamma - 1)} - (\gamma+1)(\alpha\beta+2\alpha+2\beta+\gamma-2),$$

where, as before,

$$S = 1 + \frac{\alpha \cdot \beta}{3 \cdot \gamma} + \frac{\alpha \cdot \alpha + 1 \cdot \beta \cdot \beta + 1}{3 \cdot 4 \cdot \gamma \cdot \gamma + 1} + \&c.;$$

this is the formula in hypergeometric series required for the present purpose, and it is certainly true when the series is finite.

Write now

$$\alpha = \frac{1}{2}, \quad \beta = r + 1, \quad \gamma = r - \frac{1}{2};$$

then the first term is $[1]^{r+2} \div [\frac{1}{2}]^{r+1}$, which vanishes on account of the numerator, and the second term is $-\frac{1}{2}r(r + \frac{1}{2})$, and we have consequently

$$3\Pi_1 = -\frac{1}{2}r(r + 2)(r + \frac{3}{2}) \cdot \frac{1}{2}S = -\frac{1}{2}r(r + \frac{1}{2}),$$

which gives

$$S = \frac{4r(r + \frac{1}{2})}{(r + 2)(r + 3)},$$

S being here the series in r , the sum of which was required, and the particular case of Mr Kirkman's formula is thus verified. It is probable that the general case might be treated in an analogous manner by first grouping together the terms which correspond to a given difference $x - y$, and ultimately summing the sums of these partial series; but I have not examined this question.

2, Stone Buildings, W.C., April 18, 1857.